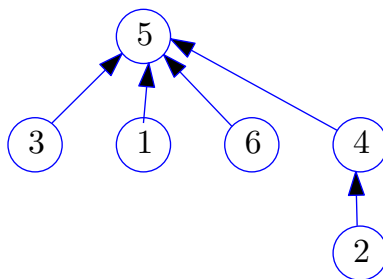


1:50-2:15pm, Monday, April 13, 2026.

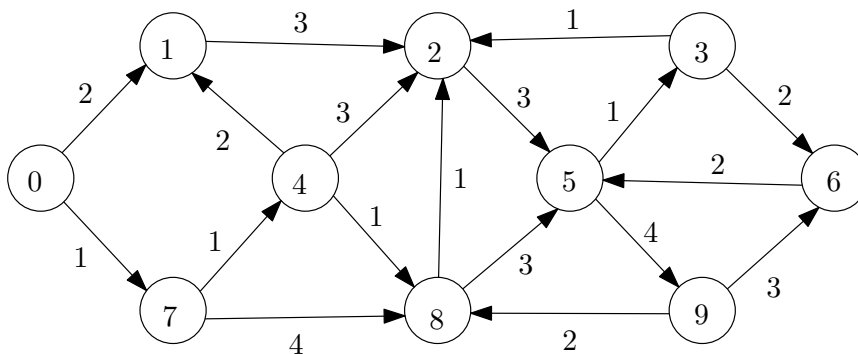
1. **Union-find data structure.** Recall that $\text{UNION}(i, j)$ merges trees containing i and j by adding an edge from the root of the smaller tree to the root of the larger tree (ties are broken arbitrarily). Assume that $S = \{1, 2, \dots, 6\}$, the initial partition consists of 1-element sets, and we are given the following sequence of operations: $\text{Union}(1, 5)$, $\text{Union}(3, 5)$, $\text{Union}(3, 6)$, $\text{Union}(2, 4)$, $\text{Union}(2, 5)$, $\text{Find}(1)$.

Answer. There are several correct solutions because of the tie-breaking rule. One solution is



In this case, we see that $\text{Find}(1) = 5$.

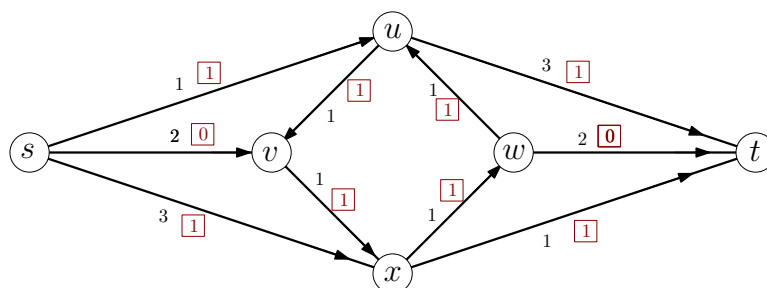
2. **Checking Correctness.** Is $D[0..9] = [0, 1, 4, 8, 3, 7, 10, 2, 4, 7]$ the array of distances from vertex 0 to vertices $0, 1, 2, \dots, 9$ in the edge-weighted directed graph below? Explain why or why not.



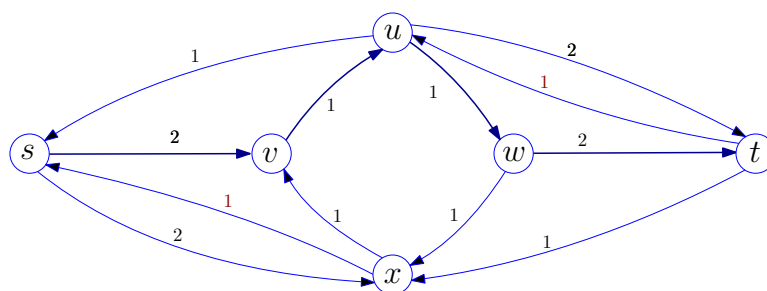
Answer. No, $D[0..9]$ are not the shortest path distances from 0. For example, for the directed edge $(0, 7)$ we have $D[7] > D[0] + w(0, 7) \Leftrightarrow 2 > 0 + 1$, which means that we can find a path from 0 to 7 of length less than 2.

(TURN THE PAGE)

3. **Network flow.** In the capacitated network below, plain numbers are edge capacities, and boxed numbers are flow values. Is this a maximum flow? Either find an augmenting st -path or find a minimum ST -cut.



Answer. The residual graph G_f below shows an augmenting st -path (s, v, u, w, t) .



4. **Vertex capacities.** We add one new constraint for the network above: The total flow entering (or leaving) vertex u cannot exceed $c(u) = \frac{3}{2}$. Draw a modified network below (with capacities, but no flow values) that models this problem.

Answer. We split the vertex u into two vertices, u_{in} and u_{out} , and add a new edge $(u_{\text{in}}, u_{\text{out}})$ with capacity $c(u_{\text{in}}, u_{\text{out}}) = \frac{3}{2}$.

